Association of binding heterogeneity of the TSPO PET ligand [^{11}C]PBR28 in deep white matter lesions and their longitudinal MRI evolution

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Introduction

- Radioligand [^{11}C]PBR28 binding enables non-invasive *in vivo* assessment of the expression of brain TSPO expression, a marker microglia and activated astrocytes
- White matter lesions (WML's) in MS display heterogeneous patterns of neuroinflammatory cells[1]:
  - "Rim" active inflammatory lesions have increased populations of activated microglia at the rim of the lesion and very few inflammatory cells in their cores.
  - "Smoldering" active inflammatory lesions which have large populations of reactive astrocytes and activated microglia and are characterized histologically by florid myelin breakdown.
  - "Burnt out" non-active inflammatory lesions which have decreased populations of inflammatory cells and have decreased axonal density.
- Inflammatory rim activation for lesions is a mechanisms for slow expansion over time
- Here we have tested whether future WML volumetric change is predicted by the TSPO PET neuroinflammatory profile at baseline.

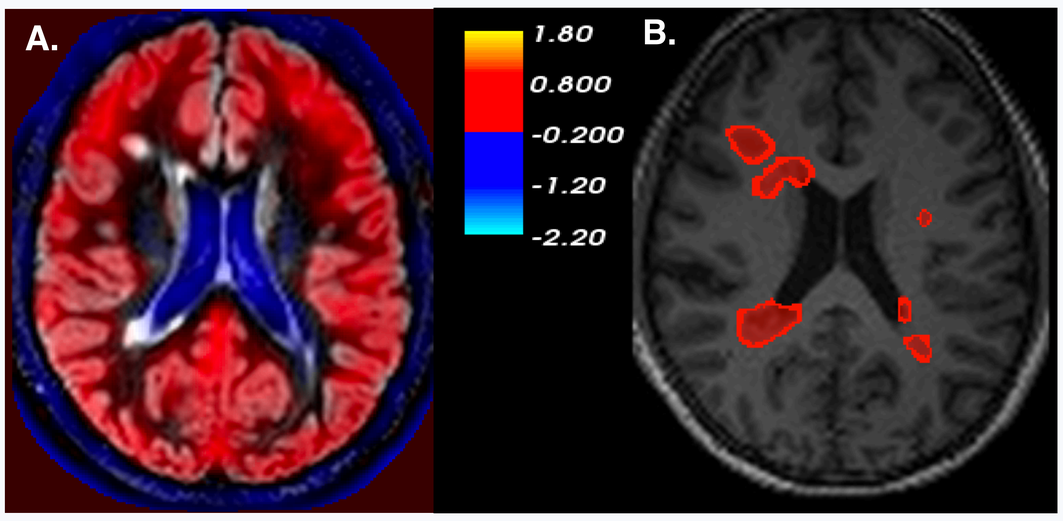


Fig 1. A, Binding potential image of MS subject. B. White matter lesion mask for the same with lesion rims that are automatically defined are shown.

Methods

- 28 MS patients and 20 healthy controls underwent PET + MR imaging with radioligand [^{11}C]PBR28
- 19 MS patients returned one year later for follow-up MR imaging.
- Pseudo-references were selected as at least 9-voxel contiguous regions of NAWM whose Vt was within the mean plus 0.25 of the standard deviation of Vt within healthy control NAWM (MAB = 2.45 ± 0.21, HAB = 4.60 ± 0.36).
- White matter lesions were segmented in baseline and one-year follow-up T2 FLAIR images.
- WML rims were determined through erosion (3 voxel depth)[2]
- WML's were categorized based on change in volume $\geq 8.5\%$
- A three Gaussian mixture model was fitted to the BP_{ND} within WML's.

	MS GM	HC GM	MS WM	HC WM
MAB	0.204 (-0.700,1.539)	0.104 (-0.781,1.389)	0.095 (-0.648,1.510)	0.036 (-0.675, 1.338)
HAB	0.257 (-0.701,1.341)	0.167 (-0.736,1.242)	0.120 (-0.656,1.326)	0.030 (-0.636, 1.164)

Table 1. Binding potential within MS and HC subjects white matter and grey matter for each binding affinity shown median (min, max).

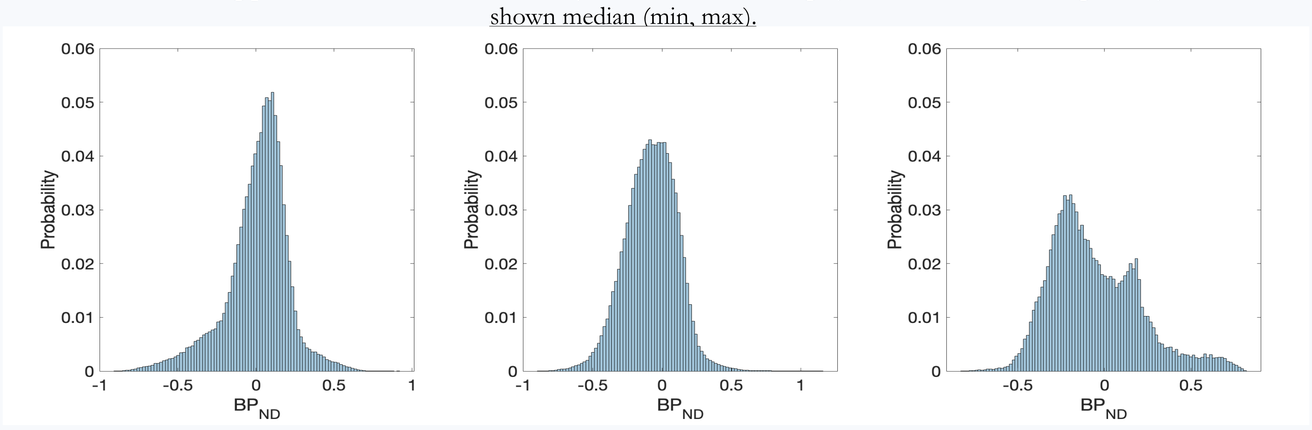


Fig 2a) Binding potential within growing WML's. 2b) Binding potential within stable WML's. 2c) Binding potential within shrinking WML's.

Results

- A significantly greater (see table 2.) rim BP_{ND} was found in growing WML's relative to shrinking and stable WML's ($p<0.01$, U-test).
- A significantly lower rim BP_{ND} was detected in shrinking WML's relative to growing and stable WML's ($p<0.01$, U-test).
- Over the cohort, 90 WML's were classified as growing, 31 as stable and 134 as shrinking by volume change over a one-year period.
- WML volumetric change over a one-year period was predicted to 56% accuracy using BP_{ND} alone.

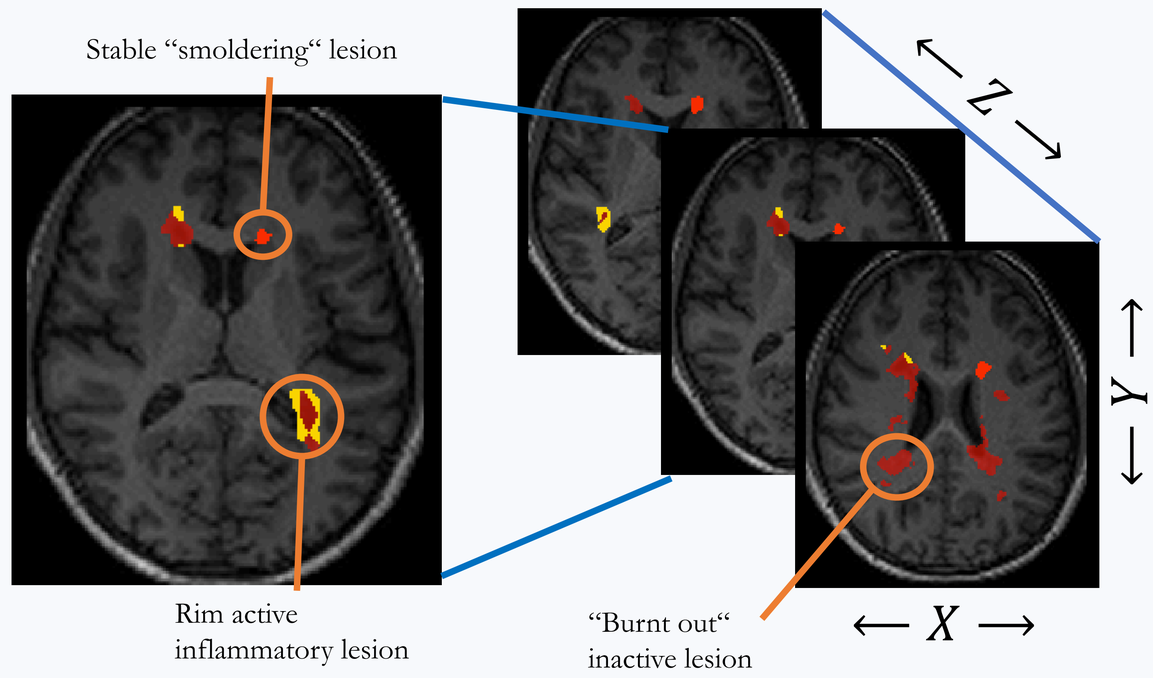


Fig 3. Spatial segmentation of the lesions within subject 24 by BP_{ND} alone using the three gaussian mixture model.

	Growing	Stable	Shrinking
Rim BPND	0.039 (-0.906, 0.911)	-0.073 (-0. 896, 1.159)	-0.105 (-0.827, 0.811)

Table 2. Global binding potential within white matter lesions of differing phenotype shown median (min, max)

	Predicted Growing	Predicted Stable	Predicted Shrinking
Actual Growing	31	0	59
Actual Stable	26	0	5
Actual Shrinking	30	0	104

Table 3. Confusion matrix of the predicted lesion type vs actual lesion type by the 3 gaussian mixture model.

Conclusion

- [^{11}C]PBR28 PET enables assessment of the spatial heterogeneity of neuroinflammation in WML in MS.
- MS WML BP_{ND} alone enables prediction of future lesion volumetric growth with moderate accuracy.

Disclosure

Sam Podmore: Nothing to disclose.
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References

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